Advanced Fire Fighting (AFF)

30 Practice Questions and Answers for STCW Exams

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Practice Set 1

Q1. What is the primary purpose of boundary cooling during a fire on a ship?

A) To minimize smoke production
B) To prevent fire spread by cooling adjacent surfaces
C) To increase the temperature inside the compartment
D) To accelerate the combustion process

Correct Answer: B (To prevent fire spread by cooling adjacent surfaces)

Explanation: Boundary cooling involves applying water to surfaces adjacent to a fire. Its primary purpose is to absorb heat, preventing transfer via conduction, convection, and radiation, thereby inhibiting the fire from spreading to new compartments or fuel sources. This maintains the structural integrity and confines the fire.

Q2. In advanced firefighting, what is the purpose of using a Class A foam?

A) To cool down hot surfaces
B) To suppress flammable gas release
C) To control oil and fuel fires
D) To neutralize acidic fumes

Correct Answer: C (To control oil and fuel fires)

Explanation: Class B foam is specifically designed for flammable liquid fires (oil, fuel, grease), forming a blanket that separates the fuel from oxygen and cools the surface. This creates a vapor seal to prevent reignition. Class A foam is generally for ordinary combustibles like wood or paper.

Q3. During a firefighting operation, what does the term "smoke curtain" refer to?

A) A physical barrier to prevent the entry of firefighters
B) A cloud of water droplets to control temperature
C) A curtain made of fire-resistant material to contain smoke
D) A warning signal for evacuation

Correct Answer: C (A curtain made of fire-resistant material to contain smoke)

Explanation: A smoke curtain, often part of a ship's passive fire protection system, is a flexible, fire-resistant barrier. Its purpose is to automatically deploy and contain smoke and hot gases, preventing their spread to other areas, thereby improving visibility and maintaining tenable conditions for escape.

Q4. Which firefighting equipment is specifically designed for use in confined spaces onboard a ship?

A) Fire hoses
B) Breathing apparatus
C) Foam monitors
D) Fire extinguishers

Correct Answer: B (Breathing apparatus)

Explanation: Breathing apparatus (BA) is essential for firefighters entering confined or smoke-filled spaces where oxygen levels may be depleted or toxic gases are present. It provides an independent supply of breathable air, ensuring the safety and operational effectiveness of personnel in hazardous atmospheres.

Q5. What is the primary purpose of the International Shore Connection (ISC) on a ship?

A) To supply freshwater to the crew
B) To connect with the shore power supply
C) To facilitate shore-based firefighting assistance
D) To enhance communication with nearby vessels

Correct Answer: C (To facilitate shore-based firefighting assistance)

Explanation: The International Shore Connection (ISC), mandated by SOLAS, is a standardized flange on ships. Its primary purpose is to connect the ship's fire main to a shore-based water supply or another vessel's fire main, enabling external firefighting assistance if the ship's pumps are insufficient or incapacitated.

Q6. In advanced firefighting, what is the purpose of using a water fog nozzle?

A) To create a dense fog for cooling purposes
B) To increase the range of water projection
C) To generate a fine mist for fire suppression
D) To create a visual signal for distress

Correct Answer: C (To generate a fine mist for fire suppression)

Explanation: A water fog nozzle disperses water into fine droplets, creating a large surface area for rapid heat absorption and conversion into steam. This rapidly cools the fire, displaces oxygen, and can protect firefighters from radiant heat, making it effective for general fire suppression and boundary cooling.

Q7. What is the role of a fire blanket in firefighting?

A) To smother small fires by cutting off the oxygen supply
B) To cool down hot surfaces in a fire-affected area
C) To create a barrier against radiant heat
D) To provide an escape route for trapped personnel

Correct Answer: A (To smother small fires by cutting off the oxygen supply)

Explanation: A fire blanket is used to smother small fires, particularly those involving cooking oil or clothing, by cutting off the oxygen supply. Made of fire-resistant material, it forms an airtight seal over the flames, interrupting the combustion process effectively and safely.

Q8. In the context of firefighting, what does the acronym PASS stand for?

A) Pressure Application Safety System
B) Portable Aqueous Suppression System
C) Pull, Aim, Squeeze, Sweep
D) Primary Air Supply Source

Correct Answer: C (Pull, Aim, Squeeze, Sweep)

Explanation: PASS is a mnemonic acronym for the correct sequence of operations when using a portable fire extinguisher: Pull the pin, Aim the nozzle at the base of the fire, Squeeze the handle, and Sweep from side to side. This ensures effective and safe extinguisher deployment.

Q9. What is the purpose of a fire screen in advanced firefighting?

A) To protect firefighters from radiant heat
B) To separate different compartments for ventilation
C) To extinguish fires through a fine mist
D) To detect flammable gases

Correct Answer: A (To protect firefighters from radiant heat)

Explanation: A fire screen, or thermal barrier, is utilized to shield firefighters from intense radiant heat emanating from a fire. It allows personnel to approach closer to the fire source for more effective direct attack, while significantly reducing the risk of heat-related injuries and fatigue.

Q10. When using a fire hose, what is the recommended nozzle pattern for initial cooling of a fire in a confined space?

A) Straight stream
B) Fog pattern
C) Solid stream
D) Cone spray

Correct Answer: B (Fog pattern)

Explanation: For initial cooling in a confined space, a fog pattern nozzle is recommended. The fine water droplets rapidly absorb heat and turn into steam, displacing oxygen and pushing back heat and smoke, thereby improving conditions for firefighters and preventing flashover.

Q11. In advanced firefighting, what is the primary function of a fire axe?

A) Ventilating smoke from the compartment
B) Cutting through bulkheads and doors
C) Spraying water in a concentrated stream
D) Applying foam to the fire

Correct Answer: B (Cutting through bulkheads and doors)

Explanation: A fire axe is a crucial tool for firefighters, primarily used for forcible entry, breaking through lightweight bulkheads, doors, or clearing obstructions to gain access to a fire. Its design enables effective penetration and leverage in emergency situations.

Q12. What does the term "fire tetrahedron" refer to in the context of firefighting?

A) A four-sided firefighting tool
B) The four elements required for combustion
C) A specialized firefighting drill
D) A geometric shape used in fire suppression

Correct Answer: B (The four elements required for combustion)

Explanation: The fire tetrahedron illustrates the four components necessary for fire: heat, fuel, oxygen, and a chemical chain reaction. Removing any one of these elements will extinguish the fire, which forms the fundamental principle behind all firefighting strategies and agent selection.

Q13. In the event of an electrical fire onboard, which firefighting equipment is specifically designed for use?

A) CO2 (Carbon Dioxide) extinguisher
B) Foam extinguisher
C) Dry powder extinguisher
D) Water mist extinguisher

Correct Answer: A (CO2 (Carbon Dioxide) extinguisher)

Explanation: CO2 (Carbon Dioxide) extinguishers are specifically designed for electrical fires (Class C). Carbon dioxide is a non-conductive gas that smothers the fire by displacing oxygen without leaving a residue, making it ideal for protecting sensitive electrical equipment.

Q14. What is the purpose of a fireman's outfit (protective clothing) in advanced firefighting?

A) To identify firefighting personnel
B) To enhance physical agility
C) To provide protection from heat and flame
D) To store firefighting tools

Correct Answer: C (To provide protection from heat and flame)

Explanation: A fireman's outfit, mandated by SOLAS, provides essential personal protective equipment (PPE) for firefighters. It is constructed from heat-resistant materials to shield the wearer from high temperatures, flames, radiant heat, and minor mechanical hazards, ensuring safety during firefighting operations.

Q15. In the context of fire safety, what does the acronym "RECAAR" stand for?

A) Rapid Escape and Control Against Radiant heat
B) Rescue, Evacuation, and Containment of Aerial Risks
C) Response, Escape, and Control Against Adverse Reactions
D) Rescue, Evacuation, Containment, Assessment, and Recovery

Correct Answer: D (Rescue, Evacuation, Containment, Assessment, and Recovery)

Explanation: RECAAR is a structured approach to emergency response and incident management, particularly in maritime contexts. It stands for Rescue, Evacuation, Containment (of the incident), Assessment (of damage/situation), and Recovery (returning to normalcy), guiding comprehensive emergency procedures.

Q16. What is the purpose of a flame arrester in firefighting?

A) To control the spread of flammable liquids
B) To arrest the progression of flames in a compartment
C) To provide visual cues during firefighting operations
D) To enhance communication among firefighters

Correct Answer: A (To control the spread of flammable liquids)

Explanation: A flame arrester is a device fitted to tank vents or pipes carrying flammable vapors. Its primary purpose is to prevent the propagation of a flame through a flammable gas mixture by rapidly cooling the flame below its ignition temperature, thus stopping flashback or explosion.

Q17. During a shipboard fire emergency, what is the primary purpose of the muster list?

A) To account for all personnel and passengers
B) To provide instructions for firefighting techniques
C) To allocate firefighting responsibilities
D) To identify the location of fire hydrants

Correct Answer: A (To account for all personnel and passengers)

Explanation: The muster list, as per SOLAS, is a crucial document detailing duties for crew members in various emergencies, including fire. Its primary purpose during an emergency muster is to ensure all personnel and passengers are accounted for and to verify their safety and assigned roles.

Q18. What is the recommended method for handling an injured person during a shipboard firefighting operation?

A) Immediate evacuation to the shore
B) Placing the person in a confined space
C) Moving the person to a designated safe area
D) Administering first aid at the fire location

Correct Answer: C (Moving the person to a designated safe area)

Explanation: During a shipboard fire, the immediate priority for an injured person is to move them quickly and safely to a designated safe area, away from direct fire hazards or smoke. Administering first aid should follow once the individual is in a secure environment.

Q19. Which firefighting agent is commonly used to suppress fires involving flammable metals?

A) Water
B) Foam
C) Dry powder
D) CO_2 (Carbon Dioxide)

Correct Answer: C (Dry powder)

Explanation: Dry powder extinguishers (Class D) are specifically designed for fires involving combustible metals like magnesium, titanium, or sodium. These specialized powders smother the fire and form a crust that excludes oxygen, effectively suppressing these unique and intense fires.

Q20. What is the purpose of a fire control plan on a ship?

A) To outline emergency escape routes for passengers
B) To specify the locations of fire alarm pull stations
C) To provide guidance for controlling and suppressing fires
D) To identify the ship's navigation routes

Correct Answer: C (To provide guidance for controlling and suppressing fires)

Explanation: The fire control plan, prominently displayed on ships as per SOLAS, provides critical information on the ship's fire protection arrangements. It guides effective fire control and suppression by indicating locations of fire-extinguishing appliances, fire-resistant divisions, and means of escape.

Q21. What is the purpose of a deluge system in shipboard firefighting?

A) To deliver a high-pressure water mist
B) To flood a compartment with water rapidly
C) To control electrical fires with a dry powder
D) To generate foam for flammable liquid fires

Correct Answer: B (To flood a compartment with water rapidly)

Explanation: A deluge system is a fixed fire suppression system designed to rapidly discharge a large volume of water over a protected area or into a compartment. It's used in high-hazard areas to quickly cool and smother large fires or create an effective water barrier.

Q22. During a shipboard firefighting operation, what is the primary function of a portable gas detector?

A) To locate fire hydrants
B) To identify flammable gases
C) To measure ambient temperature
D) To monitor radio communication

Correct Answer: B (To identify flammable gases)

Explanation: A portable gas detector is a vital piece of equipment used to monitor the atmosphere for the presence of flammable, toxic, or oxygen-deficient gases. It ensures the safety of personnel entering confined spaces or areas potentially affected by gas leaks during a fire emergency.

Q23. In the context of advanced firefighting, what is the purpose of a water curtain?

A) To create a barrier against radiant heat
B) To supply water to firefighting hoses
C) To generate foam for liquid fires
D) To create a visual signal for distress

Correct Answer: A (To create a barrier against radiant heat)

Explanation: A water curtain, formed by a series of nozzles spraying water, creates a dense sheet of water. This acts as a thermal barrier, absorbing radiant heat and protecting adjacent areas or personnel from intense heat radiation, thereby preventing fire spread or providing a safe egress path.

Q24. Which firefighting equipment is commonly used to cut ventilation openings in ship compartments?

A) Fire axe
B) Water fog nozzle
C) Thermal imaging camera
D) Hydraulic rescue tool

Correct Answer: D (Hydraulic rescue tool)

Explanation: Hydraulic rescue tools, such as spreaders and cutters, are powerful implements used for heavy-duty cutting and forcing open structural components. They are highly effective for cutting ventilation openings in robust ship compartments or structures where conventional tools are insufficient.

Q25. During a shipboard firefighting operation, what is the primary purpose of a positive pressure breathing apparatus (PPBA)?

A) To supply fresh air to the compartment
B) To prevent smoke from entering the mask
C) To maintain positive pressure inside the mask
D) To enhance communication among firefighters

Correct Answer: C (To maintain positive pressure inside the mask)

Explanation: A positive pressure breathing apparatus (PPBA) maintains a slightly higher pressure inside the mask than the ambient pressure. This crucial feature prevents toxic smoke or gases from leaking into the mask, even with minor seal imperfections, ensuring the wearer breathes only clean, supplied air.

Q30. In advanced firefighting, what is the primary purpose of a fire-resistant bulkhead?

A) To provide a visual barrier against smoke
B) To contain the spread of fire within compartments
C) To enhance communication among firefighting teams
D) To store firefighting equipment safely

Correct Answer: B (To contain the spread of fire within compartments)

Explanation: Fire-resistant bulkheads are structural divisions designed to withstand fire for a specified period (e.g., A-60, B-15). Their primary purpose, as per SOLAS, is to contain a fire within its compartment of origin, preventing its spread to other parts of the ship and protecting escape routes.

Practice Set 2

Q1. A safety is provided on the CO2 discharge piping to prevent:

A) Flooding of a space where personnel are present
B) Over pressurization of the CO2 discharge piping
C) Rupture of cylinder due to temperature increase
D) Over pressurization of the space being flooded

Correct Answer: B (Over pressurization of the CO2 discharge piping)

Explanation: A safety relief device, often a bursting disc, is installed on CO2 discharge piping to prevent over-pressurization. In the event of an abnormal pressure build-up, it ruptures to release the CO2, protecting the piping system from damage or catastrophic failure, compliant with the FSS Code.

Q2. Apart from red what other color is frequently used for a CO2 extinguisher:

A) Blue
B) Yellow
C) Black
D) Red

Correct Answer: C (Black)

Explanation: While the main body of fire extinguishers is typically red, different types are identified by a color band or the entire body color. For CO2 extinguishers, the body is often black or has a black band, clearly distinguishing them from other types for rapid identification.

Q3. Flue gases are used directly for inerting?

A) True
B) False

Correct Answer: B (False)

Explanation: False. Flue gases are not used directly for inerting. They must first be processed by an inert gas generator system, which cleans, cools, and removes sulfur oxides and particulate matter, before being supplied to cargo tanks for inerting purposes to prevent explosions.

Q4. In fighting a fire in a fuel tank, the first action you should attempt is to:

A) Station someone at the fixed CO_2 release controls

B) Top off the tank to force out all vapors

C) Begin transferring the fuel to other tanks

D) Secure all sources of fresh air to the tank

Correct Answer: D (Secure all sources of fresh air to the tank)

Explanation: For a fire in a fuel tank, the first critical step is to eliminate the oxygen supply by securing all openings, vents, and fresh air sources. This starves the fire, helping to contain it before applying extinguishing agents and mitigating the risk of rapid spread.